

pwned: exposure from data breaches

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Research revision: September 10, 2026

the price of a data breach

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Yahoo! revises the estimate for the latter breach to 3 billion.

the consequence of a low price of a breach

Data
- YouGov

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- Matched sampling; comparisons below.

	CPS	YouGov	Difference
Age			
18 to 25	0.14	0.15	-0.01
26 to 35	0.18	0.18	0.00
36 to 50	0.25	0.23	0.01
51 to 65	0.25	0.26	-0.01
66 to 80+	0.19	0.18	0.01
Sex			
Male	0.48	0.49	0.00
Female	0.52	0.51	0.00

	CPS	YouGov	Difference
Race			
White alone	0.78	0.64	0.13
Black or African American alone	0.13	0.12	0.01
American Indian and Alaska Native alone	0.01	0.01	0.00
Educational Attainment			
No high school diploma	0.11	0.07	0.04
High school or equivalent	0.29	0.33	-0.05
Some college, less than 4-yr degree	0.28	0.31	-0.03
Bachelor's degree or higher	0.32	0.29	0.03

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- Information about 5,235,843,322 breached accounts

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- Have I Been Pwned (HIBP):
 - Clearinghouse of information about data breaches
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- Query HIBP with the email ID on the file.

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Empirical Caveat

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Empirical Caveat

- Some breaches are never discovered
- We can only estimate the *lower bound*.

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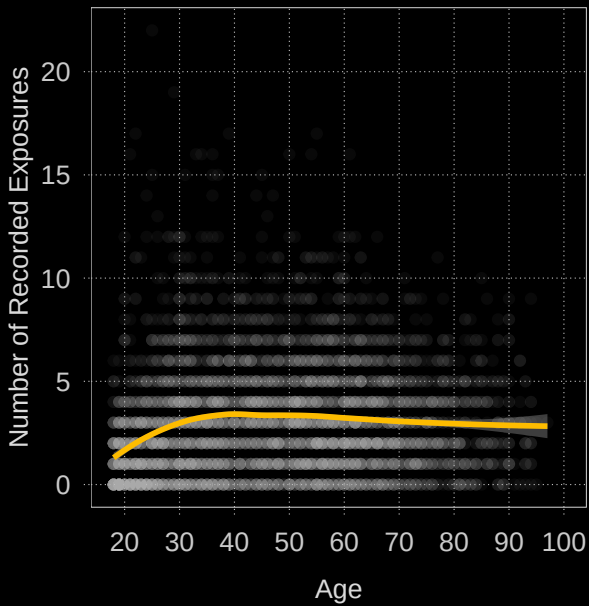
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who is pwned?

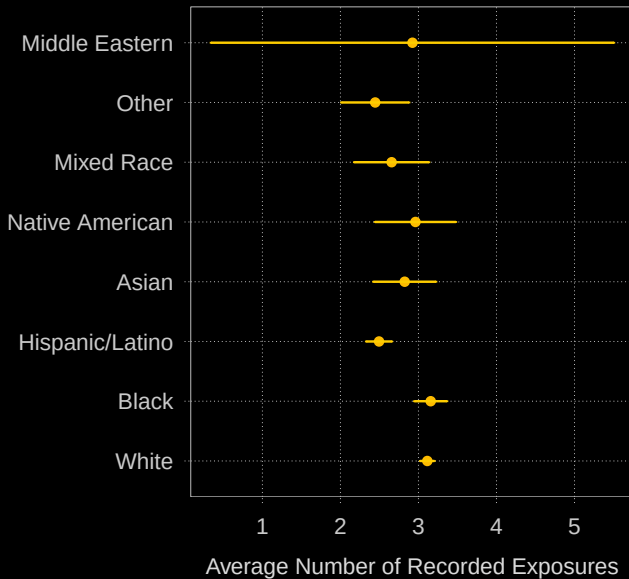
the age of the pwned



Recorded Exposures by Age

Term	Estimate	SE	Lower	Upper	p
(Intercept)	1.989	0.091	1.810	2.167	<0.001
Age spline 1	2.281	0.203	1.884	2.679	<0.001
Age spline 2	-0.900	0.209	-1.309	-0.491	<0.001

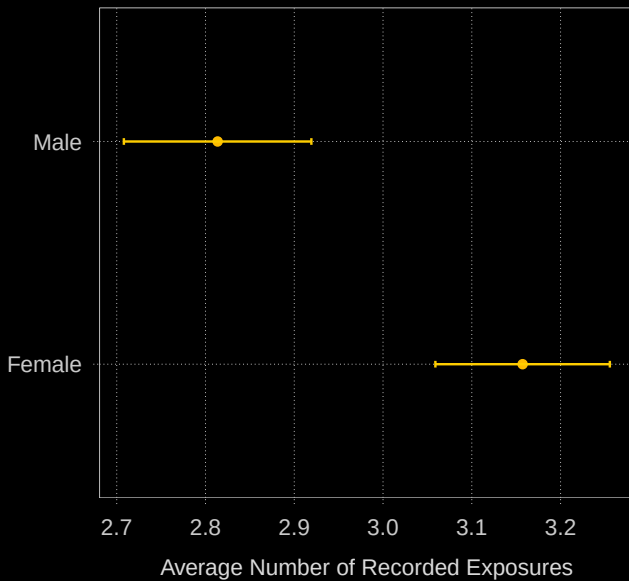
the race of the pwned



Recorded Exposures by Race/Ethnicity

Term	Estimate	SE	Lower	Upper	p
(Intercept)	3.113	0.046	3.024	3.203	<0.001
Black	0.044	0.116	-0.184	0.271	0.7069
Hispanic/Latino	-0.618	0.104	-0.822	-0.415	<0.001
Asian	-0.290	0.227	-0.735	0.155	0.2021
Native American	-0.152	0.363	-0.863	0.559	0.6757
Mixed Race	-0.457	0.234	-0.915	0.001	0.0507
Other	-0.667	0.288	-1.233	-0.102	0.0207
Middle Eastern	-0.190	0.721	-1.603	1.223	0.7919

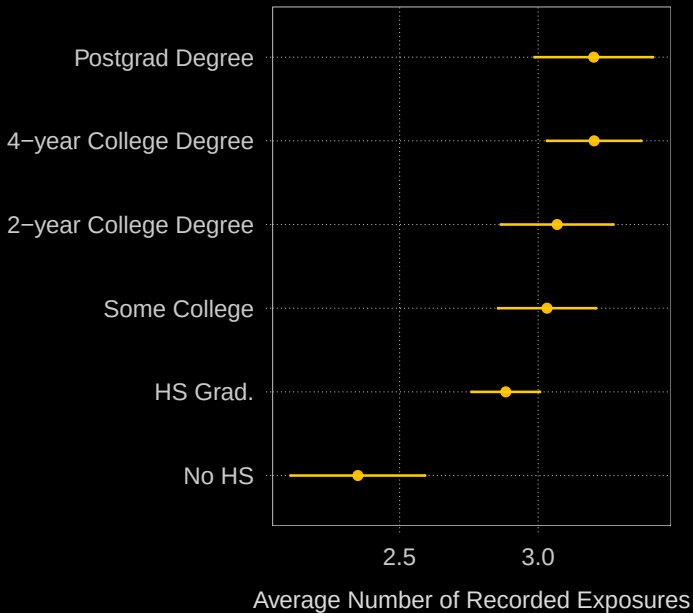
the sex of the pwned



Recorded Exposures by Sex

Term	Estimate	SE	Lower	Upper	p
(Intercept)	3.157	0.051	3.057	3.258	<0.001
Male	-0.344	0.074	-0.488	-0.199	<0.001

the education of the pwned



Recorded Exposures by Education

Term	Estimate	SE	Lower	Upper	p
(Intercept)	2.349	0.142	2.071	2.627	<0.001
HS Grad.	0.534	0.155	0.229	0.839	<0.001
Some College	0.683	0.164	0.361	1.004	<0.001
2-year College Degree	0.719	0.180	0.367	1.072	<0.001
4-year College Degree	0.853	0.166	0.528	1.178	<0.001
Postgrad Degree	0.851	0.180	0.498	1.205	<0.001

who is most at blame?

or most frequently implicated domains

Domain	Records
rivercitymediaonline.com	2911
linkedin.com	1089
modbsolutions.com	1067
myspace.com	1059
data4marketers.com	996
cashcrate.com	854
adobe.com	607
disqus.com	570
ticketfly.com	393
tumblr.com	336
dropbox.com	286
dailymotion.com	255
last.fm	244
evony.com	169
clixsense.com	150
cafemom.com	145
imesh.com	144
kickstarter.com	140
edmodo.com	128
zomato.com	112
neopets.com	106

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rivercitymediaonline.com	2911
linkedin.com	1089
modbsolutions.com	1067
myspace.com	1059
data4marketers.com	996
cashcrate.com	854
adobe.com	607
disqus.com	570
ticketfly.com	393
tumblr.com	336
dropbox.com	286
dailymotion.com	255
last.fm	244
evony.com	169
clixsense.com	150
cafemom.com	145
imesh.com	144
kickstarter.com	140
edmodo.com	128
zomato.com	112
neopets.com	106

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rivercitymediaonline.com	2911
linkedin.com	1089
modbsolutions.com	1067
myspace.com	1059
data4marketers.com	996
cashcrate.com	854
adobe.com	607
disqus.com	570
ticketfly.com	393
tumblr.com	336
dropbox.com	286
dailymotion.com	255
last.fm	244
evony.com	169
clixsense.com	150
cafemom.com	145
imesh.com	144
kickstarter.com	140
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- online-service use is a possible explanation, not measured here
- exposures are concentrated in a few sources

thank you!

Data and scripts at:
<https://github.com/themains/pwned/>

excluding spam lists

counting only verified, non-spamlist breaches.

	Mean	SE
Age		
18 to 25	1.41	0.08
26 to 35	2.25	0.08
36 to 50	2.14	0.06
51 to 65	1.88	0.05
66 to 100	1.48	0.05
Education		
No HS	1.32	0.09
HS Grad.	1.64	0.05
Some College	1.98	0.07
2-year College Degree	1.88	0.08
4-year College Degree	2.15	0.07
Postgrad Degree	2.13	0.08

	Mean	SE
Sex		
Female	1.87	0.04
Male	1.85	0.04
Race		
White	1.97	0.04
Black	1.78	0.08
Hispanic/Latino	1.51	0.06
Asian	2.00	0.18
Native American	1.71	0.18
Mixed Race	1.77	0.19
Other	1.46	0.17
Middle Eastern	2.69	1.19